

Mathematics, 2018 EC, model exam

Grade 12 · Mathematics · 2018 E.C.

Time allowed: 120 minutes

Questions: 41

Total marks: 41

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following statement defines an arithmetic progression? (1)
- A. It is a sequence in which the ratio between consecutive terms is a positive constant.
 - B. It is a sequence in which the difference between consecutive terms is a non-zero constant.
 - C. It is a sequence in which the difference between consecutive terms is a constant.
 - D. It is a sequence in which the ratio between consecutive terms is a non-zero constant.

2. The areas of two similar triangles are 144 cm^2 and 81 cm^2 . If one side of the first triangle is 6 cm, then which of the following is the length of the corresponding side of the second triangle? (1)
- A. $\frac{81}{4} \text{ cm}$
 - B. $\frac{9}{2} \text{ cm}$
 - C. $\frac{3}{2} \text{ cm}$
 - D. $\frac{9}{4} \text{ cm}$

3. What is the image of the point $(2, -3)$ when reflected about the line $y = -x$? (1)
- A. $(-3, 2)$
 - B. $(2, 3)$
 - C. $(3, -2)$
 - D. $(-2, -3)$

4. A group of 30 men participate in a survey on language skills. The number of men who know both English and Afan Oromo was equal to the number of men who know neither of these languages. The number of men who know English is 4 more than those who know Afan Oromo. How many know Afan Oromo? (1)
- A. 26
 - B. 14
 - C. 13
 - D. 11

5. Suppose the mean, the median and the mode of a certain distribution are 32, 30.5 and 29.5 respectively, then which of the following is true? (1)
- A. The distribution is positively skewed.
 - B. The Bowley's coefficient of skewness $\beta < 0$.
 - C. The Pearson's coefficient of skewness is zero.
 - D. The distribution is negatively skewed.

6. If F and f are defined functions on $[a, b]$ such that $F'(x) = f(x)$ for all x , then $\int_a^b f(x) dx$ is equal to: (1)
- A. $F'(a) - F'(b)$
 - B. $F'(b) - F'(a)$
 - C. $F(a) - F(b)$
 - D. $F(b) - F(a)$

7. The following table presents the scores of grade 12 students of a certain school in a Mathematics Exam. (1)
- | Score | 31-40 | 41-50 | 51-60 | 61-70 | 71-80 | 81-90 | 91-100 |
|----------------|-------|-------|-------|-------|-------|-------|--------|
| No of students | 10 | 9 | 15 | 12 | 10 | 4 | 2 |
- What is the middle quartile for this data?
- A. 55.5
 - B. 50.5
 - C. 60.5
 - D. 58.5

8. Which of the following is true about the graph of $f(x) = \frac{x-2}{x^2-5x+6}$? (1)
- A. $x = 2$ and $x = 3$ are its vertical asymptotes. B. The x -axis is its horizontal asymptote.
C. $y = 0$ is its oblique asymptote. D. The x -intercept is $(2, 0)$.
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9. Which one of the following is non-random sampling? (1)
- A. Stratified Sampling B. Systematic Sampling
C. Convenience Sampling D. Cluster Sampling
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10. Let A and B be two vectors given by $A = 3\mathbf{i} - 2\mathbf{j}$ and $B = 5\mathbf{i} + \frac{1}{2}\mathbf{j}$, then the vector $\frac{1}{2}A - \frac{2}{3}B =$ (1)
- A. $\frac{9}{2}\mathbf{i} - \frac{4}{3}\mathbf{j}$ B. $-\frac{11}{6}\mathbf{i} - \frac{4}{3}\mathbf{j}$
C. $\frac{11}{6}\mathbf{i} - \frac{4}{3}\mathbf{j}$ D. $-\frac{11}{6}\mathbf{i} + \frac{4}{3}\mathbf{j}$
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11. The simplified form of the expression $\frac{\sqrt{32} + \sqrt{48}}{\sqrt{8} + \sqrt{12}}$ is equal to: (1)
- A. $\sqrt{2}$ B. $\sqrt{4}$
C. 1 D. 2
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12. Tola purchased a TV stand for Birr 5000 and paid Birr 250 for transportation. If he sold the TV stand for Birr 5075, then which one of the following statement is true? (1)
- A. Tola gain 3.33%. B. Tola gain Birr 75.
C. Tola gain Birr 175. D. Tola loss 3.33%.
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13. The following table shows marks of 100 students in Mathematics and Physics. | Subject | Mean | Standard Deviation | |-----|-----|-----| | Mathematics | 20 | 2 | | Physics | 25 | 3 | Which subject shows the highest variability in marks? (1)
- A. Mathematics B. Physics
C. Both are equal D. Cannot determine
-
14. What is the coefficient of x^3y^5 in the expansion of $(2x - y)^8$? (1)
- A. 448 B. 8
C. -448 D. -8
-
15. If $Q_3 = 50$ and $Q_1 = 35$, then find the coefficient of quartile deviation. (1)
- A. $\frac{3}{17}$ B. $\frac{15}{2}$
C. $\frac{5}{17}$ D. $\frac{17}{3}$
-
16. If $f(x) = \sin x$ and $g(x) = \cos x$, then which of the following statement is NOT true about the graphs of f and g ? (1)
- A. Both functions have the same range.
B. The value of both functions can be less than -1.
C. The domain of both functions is $\mathbb{R}$.
D. They have the same period.
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17. If a frustum of a right circular cone with base radii 3 cm and 4 cm and a slant height of 6 cm is given, then what is the total surface area of this frustum? (1)
- A. $67\pi \text{ cm}^2$ B. $74\pi \text{ cm}^2$
C. $56\pi \text{ cm}^2$ D. $36\pi \text{ cm}^2$

18. Let A be a 3×3 matrix and k be scalar. Which of the following is true? (1)
- A. If B is obtained by multiplying one row of A by k , then $|B| = |A|$.
 B. If B is obtained by interchanging two rows of A , then $|B| = |A|$.
 C. If $B = kA$, then $|B| = k^3|A|$.
 D. If B is obtained by adding k times one row of A onto another row of A , then $|B| = k|A|$.
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19. The cost of an electronic device reduces at the rate of 5% per annum. If its present cost is Birr 32490, then what was its cost (in Birr) before two years? (1)
- A. 36,000
 B. 37,200
 C. 35,400
 D. 29,469
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20. A company manufactures two types of sweaters; type A and type B. It costs Birr 360 to make a type A sweater and Birr 120 to make a type B sweater. The company can make at most 300 sweaters and spend at most Birr 72,000 a day. The number of sweaters of type B cannot exceed the number of sweaters of type A by more than 100. The company makes a profit of Birr 200 for each sweater type A and Birr 120 for every sweater of type B. If x = the number of type A sweater and y = the number of type B sweater, then which one of the following standard forms of linear programming models should be used to maximize the profit? (1)
- A. Maximize $P = 200x + 120y$ subject to: $x + y \leq 300, 360x + 120y \leq 72000, y \leq x + 100, x \geq 0, y \geq 0$
 B. Maximize $P = 120x + 200y$ subject to: $x + y \leq 300, 360x + 120y \leq 72000, y \leq x + 100, x \geq 0, y \geq 0$
 C. Maximize $P = 200x + 120y$ subject to: $x + y \geq 300, 360x + 120y \geq 72000, y \geq x + 100, x \geq 0, y \geq 0$
 D. Maximize $P = 200x + 120y$ subject to: $x + y \leq 300, 360x + 120y \leq 72000, y \leq x - 100, x \geq 0, y \geq 0$
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21. If two dice are rolled, then what is the probability of getting a sum greater than 10 or doubles? (1)
- A. $\frac{4}{9}$
 B. $\frac{2}{9}$
 C. $\frac{1}{4}$
 D. $\frac{1}{9}$
-
22. Consider the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 6 & 1 & 9 \\ 4 & 0 & 5 \end{pmatrix}$. Which of the following is the $\det(A)$ and the cofactor of entry a_{32} respectively? (1)
- A. 5, 0
 B. -19, -9
 C. -19, 7
 D. 5, 9
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23. Which of the following is the partial fraction form of the rational expression $\frac{x^2+3}{x^3-x^2+x-1}$? (1)
- A. $\frac{2}{x-1} + \frac{x+1}{x^2+1}$
 B. $\frac{2}{x-1} + \frac{-x+1}{x^2+1}$
 C. $\frac{2}{x-1} - \frac{x+1}{x^2+1}$
 D. $\frac{2}{x-1} + \frac{x-1}{x^2+1}$
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24. Let $f(x) = \frac{2x-3}{x+1}$, for $x \neq -1$. Which of the following is the inverse of f ? (1)
- A. $f^{-1}(x) = \frac{x+3}{2-x}, x \neq 2$
 B. $f^{-1}(x) = \frac{x+3}{2x-3}, x \neq \frac{3}{2}$
 C. $f^{-1}(x) = \frac{3-x}{x-2}, x \neq 2$
 D. $f^{-1}(x) = \frac{x-3}{x+2}, x \neq -2$
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25. If three positive numbers are in the ratio of 1:3:5 and the sum of their squares is 875, then which of the following is the sum of the three numbers? (1)
- A. 45
 B. 75
 C. 90
 D. 99

26. A circular plate of metal is being expanded by heating. If the radius of the plate increases at a rate of 0.5 cm/sec, then which one of the following is the rate of increase of its area when the radius is 20 cm? (1)
- A. $80\pi \text{ cm}^2/\text{sec}$ B. $20\pi \text{ cm}^2/\text{sec}$
C. $10\pi \text{ cm}^2/\text{sec}$ D. $200\pi \text{ cm}^2/\text{sec}$
27. The domain and range respectively for the power function f defined by $f(x) = \frac{1}{3}x^{-\frac{6}{5}}$ are: (1)
- A. $\mathbb{R} \setminus \{0\}$ and $(0, \infty)$ B. $\mathbb{R} \setminus \{0\}$ and $\mathbb{R} \setminus \{0\}$
C. $(0, \infty)$ and $(0, \infty)$ D. $[0, \infty)$ and $[0, \infty)$
28. On selling an article for Birr 450, a dealer lost 10%. The cost price of the article is equal to: (1)
- A. Birr 405 B. Birr 500
C. Birr 550 D. Birr 454.55
29. If a man saves Birr 135 in the first year, Birr 150 in the second year, Birr 165 in the third year and so on, then in what time (in years) is his total saving be Birr 5,550? (1)
- A. 362 B. 37
C. 20 D. 360
30. Which of the following is the equation of the tangent line to the circle $x^2 + y^2 + 2x - 4y = 0$ at the point $(-2, 4)$? (1)
- A. $x - 2y + 8 = 0$ B. $x + 2y - 6 = 0$
C. $x - y - 10 = 0$ D. $x - 2y + 10 = 0$
31. If a polynomial function $f(x) = x^3 + ax^2 + bx + 3$ is divided by $x + 2$ the remainder is 15 and $x - 1$ is a factor of $f(x)$, then what are the values of a and b respectively? (1)
- A. $-6, 2$ B. $6, 2$
C. $2, 6$ D. $2, -6$
32. If Bona's income is Birr 180,000 per year and he pays 20% of this as house rent and 15% of the remainder on his child's education, then find the money left with him. (1)
- A. Birr 144,000 B. Birr 158,000
C. Birr 21,000 D. Birr 122,400
33. If the 6th and 11th terms of an arithmetic progression are 52 and 112 respectively, then which of the following is the common difference of this sequence? (1)
- A. 12 B. 15
C. 20 D. 4
34. If the first term of an arithmetic sequence is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then which of the following is the sum of the first 20 terms? (1)
- A. -1040 B. -1080
C. -1200 D. -2160
35. If in increasing geometric progression, the 6th term is 32 and its 8th term is 128, then what is the value of the common ratio? (1)
- A. -2 B. 4
C. 2 D. ± 1

36. Which of the following ordered pairs of numbers gives the coordinates of the point that divides a line segment with end points $A(-1, 5)$ and $B(5, 2)$ in the ratio of 1 : 2? (1)

A. $(3, 3)$ B. $(-3, 3)$
C. $(-4, -1)$ D. $(1, 4)$

37. What is the image of the circle $(x - 1)^2 + (y + 2)^2 = 4$ when it is rotated through $\theta = \frac{3\pi}{2}$ about the origin? (1)

A. $(x + 2)^2 + (y - 1)^2 = 4$ B. $(x - 2)^2 + (y + 1)^2 = 4$
C. $(x + 1)^2 + (y + 2)^2 = 4$ D. $(x + 2)^2 + (y + 1)^2 = 4$

38. Which of the following is a discrete data? (1)

A. Volume of water in a glass B. Number of students in the class
C. Temperature in the classroom D. Time taken to complete a test

39. Given sets $A = \{x : x^2 - 12x + 20 = 0\}$ and $B = \{x : x^2 + 5x - 14 = 0\}$, then which of the following is the relative complement of A with respect to B ? (1)

A. $\{-7\}$ B. $\{10\}$
C. $\{-7, 10\}$ D. $\{-10, 7\}$

40. Which one of the following statements is necessarily true? (1)

A. If a feasible region of a linear programming problem is unbounded, then there is no optimal value.
B. If a feasible region of a linear programming problem is bounded, then the objective function has either maximum value or minimum value.
C. If a feasible region of a linear programming problem is bounded, then the objective function must have optimal value(s).
D. At corner points of a feasible region of any linear programming problem there are always optimal values.

41. If $\int_0^k (2x - 18) dx = 0$, then which of the following is the value of k ? (1)

A. -3 B. 0
C. 9 D. 3